

Recent developments in fermentation technology: toward the next revolution in food production

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5.1 Introduction

Fermentation is an ancient technology as old as human civilization. Humans invented fermentation primarily as a way of preserving perishable food materials such as fruit, vegetable, and meat for periods of food scarcity (Steinkraus, 2004). Over the years, humans discovered empirically that fermentation not only preserves their food but converts it to foods and beverages with desirable sensory attributes. Fermented foods still make up a large proportion of the diets in the developing world and East Asia, although their importance had declined until recently in the West due to the introduction of modern food preservation techniques such as refrigeration. Yet, fermented foods including bread, sausage, and yogurt have remained essential foods in the western diet; a testament to the importance of fermentation not only as a preservation technology but also a technology for transforming food materials into products with desirable texture and flavor attributes (Terefe, 2016). Currently, there is a renewed interest in fermented foods in the West, driven mainly by the perceived health benefits of fermented foods (Hugenholtz, 2013), a perception shared by many traditional societies.

Today, food fermentation is undergoing a significant transformation. Developments in systems biology, “omics,” metabolic engineering and synthetic biology are enabling rational design of starter cultures (Shiferaw Terefe and Augustin, 2020) for sustainable production of food ingredients traditionally sourced from agricultural raw materials. These developments coupled with the significant advances in fermenter design and engineering will potentially have a profound impact on our food production system. This chapter gives an overview of fermentation process design, industrial fermentation processes, and current trends in food fermentation and the impact of advances in synthetic biology on fermentation and its implication to sustainable manufacture of food ingredients.

5.2 Fermentation process engineering

5.2.1 Introduction

Fermentation processes can be broadly divided into submerged and solid-state fermentation processes. Submerged fermentation processes are conducted in the presence of free water in a liquid media whereby the fermenting organisms grow submerged feeding on soluble substrates with total solid content

ranging from 5% to 10%. Submerged fermentation processes are by far the most common in biotechnological applications and the most advanced in terms of design and operation (Kaur and Vohra, 2013). Production of yogurt, beer, and condiments such as vinegar are conducted under submerged fermentation conditions. Solid-state fermentation on the other hand involves the cultivation of microorganisms in the absence or near absence of free water but sufficient moisture content and water activity that enables growth of the fermenting organisms (Rodríguez_Leon et al., 2013; Singhania et al., 2009), with total solid content ranging from 60% to 80%. Solid-state fermentation is more common in traditional fermentation of food products such as bread, cheese, miso, tempeh, and koji fermentation for soy sauce. Solid-state fermentation enables high substrate and product concentration, high productivity and the utilization of low-value substrates such as agri-food waste (Singhania et al., 2009). Nevertheless, the process is highly heterogeneous with gradients in temperature, gas, and matrix composition and biomass (Rodríguez_Leon et al., 2013) making the kinetic and thermodynamic characterization of the system difficult. There have been several progresses in modeling of solid-state fermentation processes, reactor design, and application studies (production of enzymes, pigments, organic acids, etc.) especially over the last 30 years (Singhania et al., 2009; Socol et al., 2017). However, industrial application of the technology so far is rare except in traditional food fermentation, composting and ensiling due to challenges related to scale-up, process monitoring and control, and availability of fermenter designs that enable efficient heat removal and mass transfer at a larger scale (Arora et al., 2018).

In general, fermentation processes are operated in batch, semibatch, or continuous mode. Most industrial fermentation including food fermentation processes are conducted in batch with a few exceptions such as the production of vinegar, baker's yeast, and the mycelial meat analogue Quorn. Continuous fermentation processes were employed by the brewing industry for a period of time using two approaches: a cascade and a tower fermentation system. The cascade system involved the use of three vessels, two for fermentation and one for biomass separation and could be operated continuously for up to 3 months with wort residence time of 19 h. However, it was discontinued due to excessive biomass production. In the tower system, the bottom part of the tower was packed with flocculated yeast while the upper part of the tower was a yeast separation zone with diameter twice that of the bottom part. The wort was introduced at the bottom of the tower and progressively fermented as it passes through the tower leaving the fermenter through the yeast separation zone. The tower process reduced the wort fermentation time from 1 week in a traditional batch fermentation process to 4–8 h. However, it suffers from a number of drawbacks such as long start-up time, the complexity of the plant, the need for a more skilled workforce, and the difficulty to match the flavor of the traditionally fermented beer. Thus, it was discontinued with the introduction of the cylinderoconical batch fermentation vessel that shortened the batch fermentation time to 48 h (Stanbury et al., 1995).

Batch operations are common in general due to the reduced risk of contaminations and cell mutations and the lower capital investment compared to an equivalent continuous process of the same volume although they suffer from lower productivity due to the time required for filling, sterilization, cooling prior to fermentation, and for emptying and cleaning after fermentation. A good compromise between a continuous and a batch process is a semibatch process which involves a continuous supply of a substrate(s) during certain periods of a batch process allowing process optimization and improvement in productivity. For instance, the fermentation may be conducted as a batch process until the limiting substrate is depleted and then a defined batch of the substrate is supplied continuously or intermittently at a specific rate required to extend the exponential growth phase. For a detailed discussion on the various fermentation operation modes, one may refer to Stanbury et al. (1995) and Williams (2002).

5.2.2 Fermentation process design

Fermentation engineering, especially with respect to submerged fermentation, is a mature science with standardized approaches for the design and optimization of fermentation processes and fermenter design. The initial step of the process design is usually the screening, identification, characterization, and selection of microbial strains for target applications. Recent developments in systems biology, metabolic engineering, and synthetic biology techniques are enabling the construction of efficient metabolic pathways for target products and their insertion in chassis organisms such as *E. coli*, reducing the efforts for screening and identification of suitable starters. This is followed by screening of growth media, kinetic studies to determine growth rate and the rate of substrate uptake and product formation via shaking flask experiments. Subsequently, laboratory scale fermentation studies are conducted to determine the optimal processing conditions for the highest titer (g/L), yield (g/g substrate), and productivity (g/L h) followed by a step-wise scale-up to pilot and commercial scales. Yield is especially important for low-value/high volume products, which include whole cells, primary metabolites and many secondary metabolites, where the cost of the substrate comprises a significant proportion of the total production cost (Villadsen et al., 2011b). During scale-up of submerged fermentation processes, factors such as specific power input (P/V), the volumetric mass transfer coefficient (K_La), shear rate, and dissolved oxygen concentration are used as scale-up parameters (Villadsen et al., 2011a). The design of submerged fermentation processes involves the use of the laboratory scale stirred tank reactor (STR) equipped with instrumentation and feedback control system for monitoring and control of parameters such as pH, temperature, stirring, and aeration rates that enables kinetic, thermodynamic, and hydrodynamic characterization of the system for rational process optimization and scale-up.

Such standardized design approach coupled with advances in fermenter design and process control has significantly benefited the biopharmaceutical and the biotechnology industries. Nevertheless, the current feedback process control systems are designed mainly to maintain processing parameters (e.g., pH, temperature...) at designed set values and are not able to make reactive adjustment to processing parameters based on real-time process performance data. As such, they are not adequate to prevent product rework (i.e., reprocessing) and at extreme situations product rejection or recall that may arise due to slight changes in raw material properties or the metabolic activities of the starter organisms (Panikuttira et al., 2018; Rathore et al., 2010). This has led to the introduction of the Process Analytical Technology (PAT) initiative, which is defined by FDA as “a system for designing, analyzing, and controlling manufacturing through timely measurements (i.e., during processing) of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality” (Junker and Wang, 2006). PAT involves identification of critical quality attributes (CQAs) of the target product and critical process parameters (CPPs) that impact them, development of analytic or monitoring technique for CQAs and CPPs (on-line, at-line, or off-line) and understanding the interdependence of CQAs and CPPs as the bases for designing appropriate feedback control system that enables real time process decisions based on analytic data (Rathore et al., 2010). One of the challenges in this respect is establishing accurate mechanistic models that describe the relationship between CPPs and CQAs as the basis for process control and optimization due to insufficient understanding of complex and dynamic biological processes (Junker and Wang, 2006; Mears et al., 2017). In this regard, the use of design of experiments (DOE) methodologies such as response surface (RSM) (Mandenius and Brundin, 2008) and hybrid modeling (mechanistic modeling coupled with data-driven modeling) (Mears et al., 2017) are suggested as possible options. The other issue is the lack of reliable in-line sensors for biomass, substrate, and product concentrations. Fluorescence spectroscopy coupled with

chemometrics (Faassen and Hitzmann, 2015), NIR spectroscopy (Grassi and Alamprese, 2018), and attenuated total reflectance-mid-infrared (ATR-MIR) spectroscopy coupled with chemometrics (Schorn-García et al., 2021) have been evaluated for this application. Clearly, in order to fully implement PAT, more research is required to develop robust models describing the relationship between CPPs and CQAs and reliable in-line sensors for real time measurement of CQAs.

In the case of solid-state fermentation processes, there are no standardized process design approaches. The design methodologies developed for submerged fermentation cannot be applied due to differences in the biochemical and hydrodynamic characteristics of the two processes (Soccol et al., 2017). Some of the challenges with scale-up of solid-state fermentation processes include the limited reproducibility of experimental results, the heterogeneity in temperature and substrate concentration that arises from heat generation during fermentation and the resulting evaporation of water, and the inadequacy of the mathematical models developed so far to describe the complex physical and biochemical changes that occur during solid-state fermentation (Rodríguez_Leon et al., 2013; Soccol et al., 2017).

5.2.3 Fermenter design

5.2.3.1 Submerged fermenters

The main types of bioreactors for submerged fermentation processes can be roughly divided into stirred tank (STR) and plug flow reactors. STRs are by far the most used because of their flexibility. These bioreactors are equipped with devices for efficient mixing and cooling of the medium, sensors for continuous monitoring of temperature, pH, dissolved oxygen, and level coupled with feedback control loops to control these parameters at the desired level. They are also equipped with feed ports for continuous supply of air and other gases, nutrients in the case of feed-batch operations, alkali and acid solutions for maintaining pH at the desired value, and antifoaming agents to maintain the level at desired value. Plug flow reactors are used only in applications such as remediation of very toxic wastewater where the complete conversion of the substrate is required in which case a STR is used for pretreatment followed by a plug flow reactor. There are also some variations of the stirred tank fermenter including tower fermenter, airlift fermenter, bubble column reactor and fluidized bed reactor which are pneumatically agitated instead of mechanical agitation. These fermenters are industrially used in applications such as mold fermentation where mechanical agitation induced shear stress may cause cell damage. For instance, the airlift fermenter is used in the production of Quorn, the mycelial meat analogue that is made from the biomass of *Fusarium graminearum*, which is cultivated in a continuous fermentation process (Kaur and Vohra, 2013). For more detailed discussion on submerged fermenters, one may refer to Williams (2002).

5.2.3.2 Solid-state fermenters

There are four types of commonly used solid-state fermenters; tray, packed bed, horizontal drum, and fluidized bed reactors, which vary based on the aeration and agitation strategies (Berovic, 2019) (Fig. 5.1). Tray fermenters are the most basic solid-state fermenters that are commonly used in traditional solid-state fermentation of foods in Asia where the substrates are thinly spread (5–10 cm) over trays made of stainless steel, plastic, or wood, which are stacked in a shelf on top of one another and kept in a room or a cabinet with continuous air circulation to maintain constant temperature and humidity (Fig. 5.2). The trays are typically perforated at the bottom and on the sides to facilitate air flow (Soccol et al., 2017; Berovic, 2019). Most of the other solid-state fermenters are still at the experimental

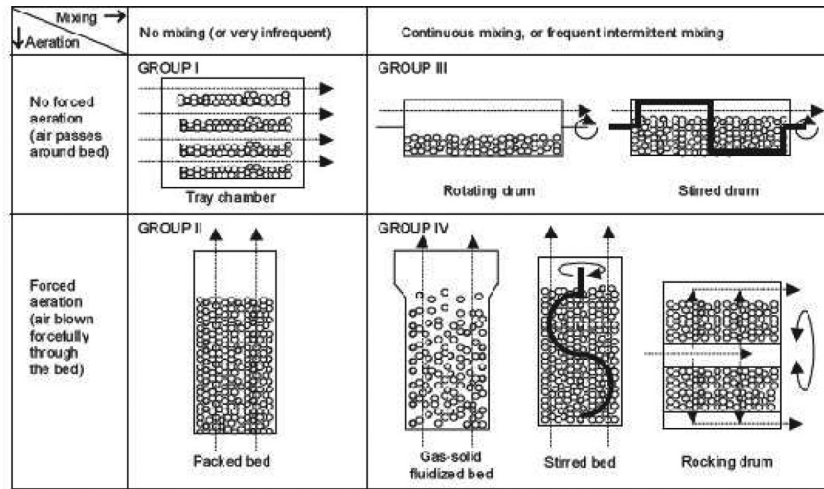


FIG. 5.1 The classification of solid-state fermenters design based on agitation and aeration strategies.

Adapted with permission from [Mitchell et al. \(2000\)](#).

stage except the drum fermenter, which has been developed by the Japanese firm Fujiwara and used for koji fermentation in many parts of Asia ([Soccol et al., 2017](#)). The drum fermenter typically consists of a cylindrical drum lying horizontally, which is operated partially filled with a bed of the substrate and air blown over the headspace. In rotary drum fermenters, the drum rotates around its central axis to mix the bed. In stirred drum fermenters, paddles mounted on a shaft along the central axis of the drum rotate and mix the bed while the drum body remains stationary ([Fig. 5.3](#)) ([Mitchell et al., 2006](#)).

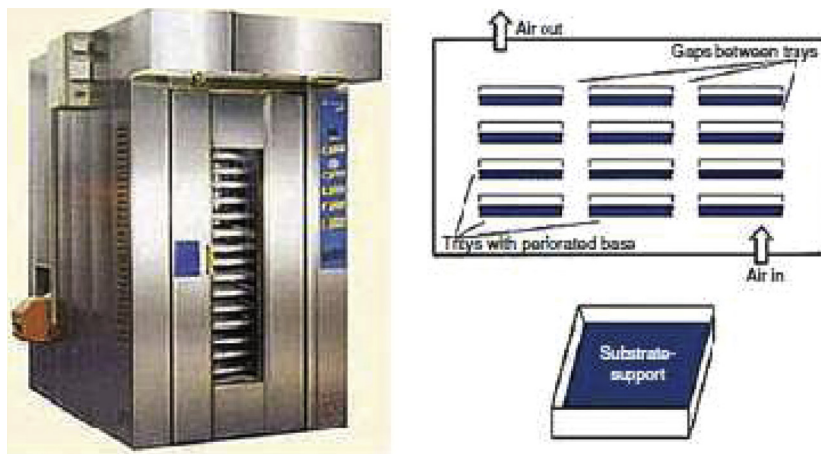
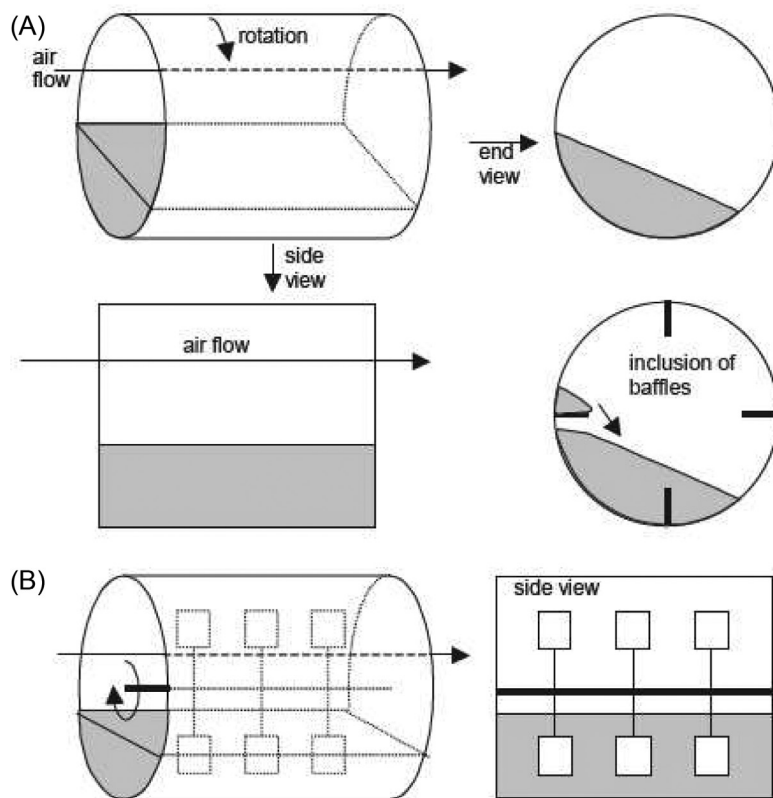


FIG. 5.2 A tray fermenter and schematic diagrams showing a group of trays and an individual tray.

Adapted with permission from [Berovic \(2019\)](#).

**FIG. 5.3**

Schematic diagrams showing the basic features of (A) rotary drum solid-state fermenter and (B) stirred drum solid-state fermenter.

Adapted with permission from Mitchell et al. (2006).

5.3 Industrial food fermentation

Food fermentation processes have significantly benefited from developments in microbiology, biotechnology and process engineering over the last century. The production of several fermented foods and beverages including wine, beer, and spirits as well as bread, sausages, sauerkraut, pickles, yogurt, and cheese have been industrialized. However, the design and operation of industrial food fermentation processes, except those for food ingredients and processing aids such as flavorings and enzymes, differ from that of standard submerged fermentation processes described in the previous sections.

5.3.1 Advances in industrial vegetable fermentation

The industrial production of fermented vegetables of commercial importance including sauerkraut, fermented cucumber (pickles), and fermented olives are little different from the traditional fermentation

of these products, except the automation of some processing steps, a degree of standardization due to better understanding of the microbiology and biochemistry of the processes and the introduction of process monitoring (periodic measurement of pH, acidity, etc.). For instance, the industrial production of fermented cucumber, often referred to as pickles, is conducted in a similar way as the traditional small-scale process, with some degree of automation dependent on the scale of the operation (Franco et al., 2016). The process involves preparation of brine solution, partial filling of the open fermentation tank with brine for cushioning the cucumbers during loading, sorting, culling out, and loading of cucumbers into the fermentation tank, closing the tank with wooden boards to keep the cucumbers submerged in the brine, final salt addition to adjust the salt content to 6 to 7% which is followed by spontaneous fermentation for 3–4 weeks depending on the temperature until all the fermentable sugars are depleted (Franco et al., 2016; Fleming, 1995). The tanks are periodically sparged with air to remove carbon dioxide especially during the first 14 days of fermentation to prevent bloating (Franco et al., 2016). The brine is usually acidified with acetic acid or vinegar to pH 4.5 or slightly lower to facilitate carbon dioxide sparging and inhibit the growth of undesirable microorganisms and prevent bloating (Fleming, 1995). A small amount of calcium chloride is also added to the brine to improve the texture of the product (Franco et al., 2016; Medina-Pradas et al., 2017). The salt concentration, the pH of the brine, acidity, and sugar are periodically monitored during the various stages of fermentation (Franco et al., 2016).

In the past, open wooden or concrete tanks with flat bottoms were used as fermentation tanks. However, they are now largely replaced by fiberglass reinforced plastic tanks of similar design with capacity ranging from 15 to 20 tons. Wooden planks secured to the tanks with brackets or semicircular fiberglass covers with holes to allow free movement of brine are used to keep the cucumbers submerged inside the brine especially during the early stages of fermentation (Franco et al., 2016). The loading of fresh cucumbers into the tanks and unloading of fermented cucumbers are done manually in small-scale operations while they are largely automated in large-scale operations. Once fermentation is completed, the pickles are either stored in a tank with added salt for preservation for up to 1 year or further processed into finished products. The pickles are typically desalted by flushing with tap water and immersed in water for up to 24 h to reduce their salt content to acceptable level. This is followed by canning in a reformulated brine containing different condiments such as sugar, cinnamon, garlic, etc. depending on the target product (Franco et al., 2016).

5.3.2 Advances in other fermentation processes

As in the case of vegetable fermentation, the basic unit operations of other traditional fermentation processes such as wine (Pérez-Torrado et al., 2018) and beer (Bamforth, 2017) fermentation did not substantially change with industrialization. Perhaps, dairy fermentation has been the most affected by technological advances and industrialization (Medina-Pradas et al., 2017; Aryana and Olson, 2017). Even then, the changes are not of profound nature. Overall, advances in microbiology and biotechnology over the last century have enabled the development of pure starter cultures for the manufacture of products such as yogurt, cheese, and beer. Moreover, better understanding of the microbiology and the biochemistry of the respective fermentation processes has led to the establishment of procedures for rigorous control of raw material properties, standardized pretreatment of substrates, and control and continuous monitoring of processing conditions (e.g., temperature, pH) that improved product safety, quality, stability, and consistency. Some degree of automation of the upstream and the downstream

processes (e.g., introduction of industrial centrifuges for beer separation) and changes in the materials of construction and design of fermentation tanks (e.g., changes from earthenware or wooden tanks to stainless steel or fiberglass tanks, introduction of the cylinderoconical vessels for beer and yogurt fermentation) have been introduced with industrialization. There have also been researches toward implementation of PAT in food fermentation processes (Panikuttira et al., 2018; Schorn-García et al., 2021). However, attempts to fundamentally alter traditional processes have met with limited success. For instance, in brewing, alternative processing approaches such as the use of enzymes as alternative to malt, continuous fermentation as opposed to the traditional batch fermentation (refer to previous section), the use of immobilized cells, and synthesizing beer from an alcoholic base and flavorings were not convincing enough to be adopted by industrial brewers (Bamforth, 2017).

Overall, industrial food and beverage fermentation processes still combine science and some element of art peculiar to the specific fermented product, with the design process and degree of automation dependent on the process.

This could be due to several factors including

- i. The nature of food substrates such as milk, grains, vegetables, and fruits, which are often complex in structure and composition, unlike the substrates in most biotechnological processes which are relatively simple sterile media composed of carbon and nitrogen sources, vitamins, and minerals designed for the target process.
- ii. The complexity of the fermentation processes themselves, which often combine both catabolic biotransformation of macromolecules as well as biosynthesis of new metabolites that determine the safety, functional, and sensory attributes of the final product, unlike standard biotechnological processes designed for the biosynthesis of target primary or secondary metabolites, which are extracted and purified in downstream processing.
- iii. The legacy from traditional processes and the expected sensory attributes, which may not be replicated if pure cultures are used, the substrate is sterilized or the process is fully automated, which necessitate a combination of both art and scientific approaches.

5.4 Recent developments in food fermentation

The last decade has seen a significant surge of interest in fermented foods and fermentation technology both by consumers and the food industry. The main drivers being the perceived health benefits of fermented foods, sustainability and consumer's increasing interest in "natural" and preservative free products, and changes in the dietary pattern of consumers, such as the rising prevalence of veganism, vegetarianism, and flexitarianism (Hugenholtz, 2013; Shiferaw Terefe and Augustin, 2020), which are fueling the plant-based trend. The world economic forum in November 2020 described fermentation as "an opportunity to fundamentally change the way the world eats and improve global human and environmental health and the economy" (GFI, 2020). Fermented foods are positioned at the center of mega trends such as healthy living (e.g., natural, clean label, functional...), ethical living (e.g., plant-based, sustainable...), and shifting market frontiers (e.g., exoticism: growing experimentation with exotic flavors especially by millennials ...) that are shaping consumer markets worldwide (International, 2018; Askew, 2018). Among growth drivers, gut health accounted for the highest proportion (1.0 relative score), followed by sugar reduction (0.22), prebiotic/probiotic delivery (0.15), and the plant-based trend (0.07) in that order. Nevertheless, interest in plant based foods increased by 532.8%

compared to 110.3% for gut health foods over the last 4 years (2016–2020) (Ravi, 2020) and it is likely that the plant-based trend will overtake gut health as the main driver for growth in fermentation over the coming years.

5.4.1 Innovations in traditional or “natural” food fermentation

The global fermented food and drink market including fermented alcoholic and nonalcoholic beverages (e.g., wine, beer, kombucha), fermented dairy (yogurt, cheese, kefir), bakery (e.g., sourdough bread, yeast raised doughnuts), and other fermented food products (e.g., sauerkraut, kimchi, pickles) was \$1036.38 billion in 2017 and is estimated to reach \$1376.42 billion by 2022 growing at a compound annual growth rate (CAGR) of 5.84% (Technavio.com, 2017). Most of the product innovation over the last 10 years has been on plant-based fermented food products. According to Dupont (Dupont, 2019), the growth in the launch of fermented dairy alternatives between 2011 and 2018 was 43% compared to 16% in fermented dairy products.

The growing interest in fermented foods has led to popularization and commercialization of traditional fermented products such as kefir and kombucha out of their home regions with a rapid expansion in their market size. For instance, the global kefir (a fermented dairy product originally from the Caucasus) market tripled between 2011 and 2016 as it became popular worldwide (Technavio.com, 2017). The global kefir market was valued at \$1.23 billion in 2019 and is forecasted to reach 1.84 billion by 2027 growing at a CAGR of 5.4% (Insight, 2020a). Similarly, the global market size of kombucha (a fermented sweet tea product originally from China), grew from \$0.5 billion to \$1.84 billion in 2019, and is forecasted to reach \$10.45 billion by 2027 growing at a CAGR of 23.2% (MarkestandMarkets, 2015; Insights, 2020b). The starter cultures of these traditional products are also being increasingly used for development of novel products. Examples include Bionade: fruit flavored malt beverages fermented using gluconic acid bacteria, one of the bacteria in the starter culture of kombucha, Rythem, coconut milk-based beverages with or without fruit juice fermented using kefir starter cultures (kefir grains), and kefir water products with various fruit juice flavors fermented using kefir grains (e.g., TheGoodSeed, Nexba). Traditional fermentation is also being employed for developing novel food ingredients and products with desirable technological or health functionality, enrichment with bioactive compounds or reducing the level of undesirable components such as sugar, undesirable flavor compounds or antinutrients (Shiferaw Terefe and Augustin, 2020). Recently, novel fermentation-based technologies were developed to enhance the yield of the bioactive isothiocyanates in brassica products (Augustin and Terefe, 2019; Cai et al., 2020), reduce sugar levels in sugar rich plant-based products via conversion into oligo- and polysaccharides and sugar alcohols (Augustin et al., 2019; Xu et al., 2020), reduction of bitter flavor notes in coffee, (Watson, 2016c) and production of a pea protein hydrolysate ingredient for enhancing salty taste of food products (Bhomik et al., 2017).

From technological innovation perspective, the application of nonthermal processing technologies such as ultrasound, high pressure (HPP) and pulsed electric field (PEF) processing are being explored for enhancing the efficiency of food fermentation processes and modulate the nutritional and sensory attributes of fermented foods with promising outcomes (Shokri et al., 2021). For instance, ultrasonic pretreatment was reported to increase the growth kinetics and metabolic activities of fermentation starter cultures (Shokri et al., 2020, 2021; Ojha et al., 2018), enhance lactose hydrolysis and conversion to galacto-oligosaccharides during milk fermentation (Nguyen et al., 2009), and shorten the fermentation time of sweet whey (Barukčić et al., 2015), and enhance the bioconversion of isoflavone

glycosides into aglycones during fermentation of prebiotic supplemented soymilk (Yeo and Liong, 2011). Similarly, HPP at sublethal level (<50 MPa) has been shown to increase the fermentative activities of starter microorganisms (Gálvez et al., 2011; Mota et al., 2018; Ferreira et al., 2019). On the other hand, HPP at higher pressures (>100 MPa) enhanced the ability of nonsaccharomyces yeast to outcompete grape wild yeast and dominate wine fermentation (Bañuelos et al., 2016), improved the acidification rate of milk during fermentation, lowered coagulation pH and improved gel strength (Tsevdou et al., 2013) and improved the texture and reduced syneresis of low-fat set yogurt (Harte et al., 2003). PEF processing has also been shown to stimulate growth and metabolic activities of fermentation starter cultures (Grosse et al., 1988; Fologea et al., 1998; Ohba et al., 2016), modulate the production of metabolites during kimchi fermentation (Joo et al., 2013), increase the production of bioactive aglycones during fermentation of mannitol supplemented soymilk (Yeo and Liong, 2013) and enhance the fermentation rate of apple cider and reduce ethanol content (Al Daccache et al., 2020). Indeed, all these technologies have potentials to enhance fermentation rate and reduce fermentation time and modulate metabolite profile enabling the manufacturing of fermented foods and beverages with differentiated sensory and nutritional attributes. However, the results so far are based on laboratory scale investigations and further research is required to integrate these technologies into standard fermentation processes at larger scale without compromising the robustness of the fermentation processes (Shokri et al., 2021).

5.4.2 Precision fermentation for production of food products ingredients

A pivotal development in food fermentation is the emergence of precision fermentation as a technology for sustainable manufacture of food ingredients that were traditionally derived from plant or animal sources as an offshoot of developments in synthetic biology. Precision fermentation is the use of genetically engineered microorganisms as starters for production of target molecules via fermentation. It is increasingly being used for the manufacture of a variety of food ingredients that were conventionally sourced from animals and plants. Examples of such products include whey proteins by Perfect Day (Watson, 2016b), production of gelatin by Geltor (Watson, 2016a), vanilla, saffron, and steviol glycosides by Evolva-Cargil (Watson, 2015, 2018), and human breast milk lactoferrin by CONAGEN (GlobeNewswire, 2020).

Precision fermentation per se is not new and genetically engineered microorganisms (recombinant DNA technology) have been used since the 1970s for fermentative production of enzymes and other biomolecules for various applications (Katz et al., 2018; Teng et al., 2021). A classic example is chymosin, that has been produced by recombinant DNA technology since the 1980s for use in cheese manufacturing as an alternative to rennet traditionally derived from calf stomach (Teng et al., 2021). Synthetic biology on the other hand has developed over the last two decades as an extension of genetic engineering by incorporating engineering design principles such as standardization, abstraction of modular parts and decoupling of design from fabrication into genetic engineering to streamline the design, build and test of synthetic gene networks and facilitate the construction of biological systems for target applications (Katz et al., 2018; Agapakis, 2014). Synthetic biology is a multidisciplinary approach that encompasses genetic programming, pathway engineering, metabolic engineering, DNA synthesis and assembly and computer aided design among other disciplines (Katz et al., 2018). It goes beyond the mere alteration of certain genes as in genetic engineering to the creation of new genetic constructs that fulfill target functionality (Boldt and Müller, 2008). The current expansion in the application of precision fermentation to the production of diverse food ingredients is a direct outcome of advances in synthetic biology, which was

facilitated by reduction in DNA sequencing and synthesis cost and development of genome editing tools such as CRISPR/Cas9 (Katz et al., 2018). Further advances would enable progress toward the design of “synthetic cell factories where all available resources are diverted to produce the desired compound” with minimal formation of by-products (Teng et al., 2021). In fact, synthetic biology and precision fermentation are believed to reduce our dependence on traditional agriculture and animal husbandry, which are constrained due to limited arable land, climate change and population growth, and enable us to sustainably feed the world and ensure food security (Teng et al., 2021; Lv et al., 2021). Nevertheless, realizing that goal requires more than an efficient cell factory. Rather, a multidisciplinary approach from synthetic biology through process engineering, public policy and social studies are required (Fig. 5.4).

Some of the current challenges with respect to precision fermentation for food production include:

- i. Scalability, economic feasibility and sustainability: most of the precision fermentation start-ups for food ingredient manufacturing are at an early stage and it is not clear whether they are economically feasible or sustainable when operated at scale (GFI, 2020; Wilcox, 2019).
- ii. Consumer perception: some consumers are concerned that synthetic biology may introduce new proteins and other molecules into foods that may be unsafe or allergenic. GMO labeling does not apply to precision fermentation manufactured ingredients in many jurisdictions (USA, Australia) as long as the molecules are nature equivalent, which makes it difficult for consumers to make informed choices (Wilcox, 2019). Appropriate labeling and transparency coupled with consumer education will be required to enhance the consumer trust and acceptance of precision fermentation ingredients (Lv et al., 2021).
- iii. Ethical concerns: synthetic biology in its boldest form is not a mere genetic alteration of an organism rather a creation of “a new life form” that fulfills a target vision/application, raising

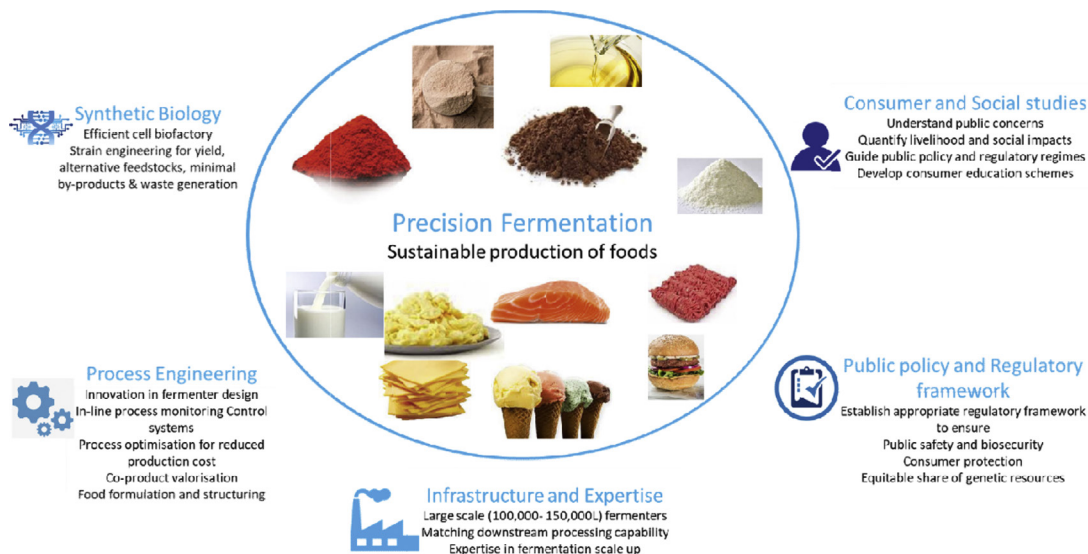


FIG. 5.4 A schematic representation of the multidisciplinary effort required to unlock the potential of precision fermentation for sustainable food production.

ethical concerns, and biosafety and biosecurity risks including escape and uncontrolled replications of these organisms to the environment and deliberate misuse by terrorists (Boldt and Müller, 2008; RRDC, 2016). In addition, manufacture of traditionally agriculture derived food ingredients like vanilla and steviol glycosides by precision fermentation affects the livelihood of farmers dependent on these commodities especially in developing countries (Lv et al., 2021; Wilcox, 2019) with societal and ethical implications. All these require concerted effort by academia, governments, and industry to ensure biosecurity, ethical use of the technology and equitable sharing of profits from genetic resources.

- iv. Regulatory aspects: policy and regulatory frameworks governing synthetic biology and precision fermentation are still being established worldwide and there are concerns that innovation in the field may outpace the development of appropriate regulation (Wilcox, 2019; RRDC, 2016).
- v. Capability in large-scale fermentation and downstream processing: Lack of expertise in fermentation process engineering at commercial scale fermentation and downstream processing facilities has been identified as the main bottlenecks for unlocking the full potential of precision fermentation. The available food fermentation capacity will be consumed in 12–24 months (Warner, 2021), highlighting the need for investment on fermentation facilities.

Overall, more research is required on discovery of better chassis organisms, design of better genetic componentry and artificial intelligence guided metabolic and pathway engineering for higher yield, titer and productivity of target molecules and reduction in by-products, strain engineering for use of alternative feedstocks other than refined sugar including agri-food waste and CO₂, innovations in bioreactors optimized for food grade fermentation, innovation in downstream processing to reduce cost, and co-product valorization, that is, value adding to spent media from fermentation processes (GFI, 2020; Lv et al., 2021) so as to ensure the economic feasibility and sustainability of precision fermentation processes. Further innovations in continuous fermentation processes and the use of immobilized microbial cells may also contribute toward ensuring the cost effectiveness of precision fermentation processes.

5.4.3 Fermentation for valorization of food waste

A significant amount of organic waste is generated in the agri-food supply chain from on-farm production through postharvest handling, processing, distribution, and consumption. Edible food waste alone accounts for over one third of the edible food produced globally, equivalent to about 1.3 billion tons (Uçkun Kiran et al., 2014). In Australia, 7.6 million tons of food waste are generated per year. Approximately 70% of this waste is edible and the cost to the Australian economy is estimated at 36.6 billion per year, which is quite substantial (Li et al., 2014). Currently, most food waste streams worldwide are used for low-value animal feed, compost, anaerobic digestion and incineration with other combustible municipal waste for energy generation (Lin et al., 2014). Clearly, food waste is a valuable resource that needs to be minimized and the unavoidable part converted to food ingredients and other valuable food products (Terefe, 2016). In this regard, fermentation is being explored as one of the enabling technologies for the conversion of food waste into high-value food ingredients and other value-added products (Lin et al., 2014; Koutinas et al., 2014). Some of the applications investigated include fermentation pretreatment of fruit processing waste for enhancing the extractability and bioactivity of phenolic antioxidants (Martinez-Avila et al., 2014; Madeira Junior et al., 2015), oil pressing cakes, fruit, vegetable, and bakery wastes as feedstocks for production of food flavorings (Laufenberg et al., 2004) enzymes (Uçkun Kiran et al., 2014), organic acids (Zhang et al., 2013), and microbial biomass

(Stabnikova et al., 2005; Di Donato et al., 2014). The food industry is also progressing toward integrated processing for efficient utilization of raw materials. An interesting example in Australia is the Manildra group that produces gluten, starch, and glucose syrup from wheat grain and ferment the waste generated in the manufacture of the three products to ethanol for food and beverage, pharmaceutical, and personal care applications enabling 100% utilization of the wheat grain (Manildra, 2021). With respect to food waste valorization, biomass fermentation, i.e. production of microbial biomass using food waste as feedstock is particularly interesting as it enables the efficient conversion of a variety of agri-food waste into high protein microbial biomass for food application (Teng et al., 2021).

5.4.4 Fermentation and the alternative protein trend

Fermentation is considered as one of the key three technologies that are enabling the alternative protein revolution with the other two being plant-based proteins and cultivated proteins (GFI, 2020). In the context of the alternative protein industry, fermentation is being explored for:

- i. Transforming plant materials into alternative protein food products like Tempeh, i.e. traditional fermentation
- ii. Cultivating microorganisms for use as a primary protein source, i.e. biomass fermentation as in the case of Quorn or
- iii. Producing specialized ingredients such as proteins, fats, enzymes, and flavoring for use in plant-based products and cultivated meat mainly via precision fermentation, an example being soy leghemoglobin that is being used in the plant based “Impossible” burger.

The focus in this respect is progressively shifting toward precision fermentation. Of the 14 newly launched alternative protein start-ups in 2019, eight were on biomass fermentation whereas of the 13 new start-ups in 2020, nine were on precision fermentation, three on biomass, and one on traditional fermentation. Fermentation companies in the alternative protein sector raised a total of \$587 million investment in 2020, which was a twofold increase from the previous year (GFI, 2020). Clearly, there is a huge market driven interest in fermentation technology for manufacture of ingredients for the alternative protein sector. Nevertheless, most of the start-ups are at a very early stage and they face most of the challenges highlighted earlier for precision fermentation including cost-effective production of the protein ingredients and therefore cost competitiveness with agriculture derived ingredients, regulatory hurdles, and limited availability of large-scale fermentation and downstream processing infrastructure and trained manpower. Perhaps, the biomass fermentation start-ups may have less hurdles in terms of economic feasibility and regulatory approval compared to the precision fermentation-based ones.

In fact, some of the biomass fermentation start-ups (e.g., Air Protein, Solar Foods) are based on the cultivation of hydrogenotrophs on potentially readily available feedstocks (carbon dioxide and hydrogen) (Linder, 2019; Alloul et al., 2021), and may contribute to greenhouse gas emission reduction.

5.5 Conclusion and future perspectives

The 20th century has witnessed the transition of fermentation from a technology for household preservation of primary produce to a sophisticated technology for manufacture of pharmaceuticals, biochemicals, food ingredients, foods, and beverages at industrial scales at the back of advances in microbiology, biotechnology and process engineering.

Fermentation is currently experiencing a resurgence and transformation. This is propelled by changes in consumer attitudes and dietary patterns, concern for the environment, sustainability, and food security on one hand and scientific progress in systems biology and metabolic engineering spawning precision fermentation enabling “designed” microbial cell factories that are able to produce biomolecules of interest using nature’s genetic blueprint. Consumer’s growing awareness of the relationship between diet and health and interest in “natural,” clean label’ and sustainably and ethically sourced food products is driving growth and innovation in traditional/natural fermented foods and beverages.

The plant-based trend, which is being fueled by similar issues, is driving growth especially in precision fermentation. Fermentation is currently in a growth trajectory that is set to stay for the foreseeable future. Indeed, precision fermentation, in particular, has a huge potential to improve sustainability, food security, human and environmental health, and the global economy. Nevertheless, further research is required to ensure the economic feasibility and scalability of precision fermentation processes including metabolic engineering and strain optimization for productivity and use of alternative feedstocks, innovations in fermenter and downstream equipment design, real-time process and product monitoring, and control system design and co-product valorization. Social science research is also needed in parallel in order to understand the ethical and social dimensions of the issues surrounding synthetic biology and precision fermentation and guide government policy and regulatory framework to ensure biosecurity and biosafety, consumer protection, and equitable share of economic dividends from genetic resources. Investment on large-scale fermentation and downstream facilities dedicated to food ingredient manufacturing is also urgently needed since lack of infrastructure and expertise at large-scale fermentation is becoming an impediment to unlocking the full potential of precision fermentation processes.

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